

Capstone Project Proposal

Team Apex Innovators | Secure, Deployable AI Tooling for KPC Operations

Inuka Hackathon — Problem 10, Domain E: HSE, Compliance & Deployment Readiness

1. Team Members & Roles

Name	Role	Primary Responsibilities
Stanley Metone	Project Lead / Data Scientist	Overall delivery, stakeholder communication, and design of the alerting/recommendation logic that turns cleaned data into actionable insight (Team Lead, self-designated by team).
David Kimathi	Data Engineer	Real-time / event-driven ingestion architecture across depot and SCADA-style feeds, automated data-quality gates, and pipeline reliability.
Jackline Mboya	ML / Data Privacy Engineer	Anonymization policy design (field masking, k-anonymity enforcement), feature pipeline for the predictive-alert model, and model evaluation.
Patrick K. Kariuki	Business Analyst	Cost-avoidance and ROI modeling, adoption-feasibility case, success-metric definition, and translating pipeline health into an executive narrative.
Abdirahim Osman	QA & Deployment Lead	CI/CD pipeline hardening (test gates, security scanning, canary deploys), monitoring/alerting setup, and deployment/handover documentation.

2. Problem Statement

Template (Module 4)

[Operational area] is experiencing [problem], which causes [business impact], measured by [current metric]. If unresolved, this will result in [consequence]. Success looks like [target metric/state].

Applied

KPC's operational data teams — spanning depot, SCADA, and maintenance-log sources — are experiencing messy, sensitive data with no consistent, automated way to clean, anonymize, and ship it into production analytics tools, which causes compliance exposure, delayed operational insight, and significant analyst time lost to manual reconciliation, measured by manual-cleanup hours per week and the absence of an auditable anonymization trail. If unresolved, this will continue to expose KPC to data-sensitivity and audit risk, slow decision-making for operations leadership, and keep useful AI tooling stuck at the prototype stage rather than reaching production. Success looks like a hardened, monitored, near-real-time data pipeline that ingests messy operational data, automatically cleans and anonymizes it to KPC's data-sensitivity rules, ships it through a tested CI/CD pipeline, and surfaces both operator-facing alerts and an executive-facing view — backed by a quantified, stakeholder-validated case for adoption beyond the hackathon.

How This Addresses the Evaluation Criteria

- **Business/Social Value:** yes — KPC and Em-Tech currently have no repeatable way to move sensitive operational data into analytics safely; this closes a real compliance and productivity gap, not a hypothetical one.

- **Technical Complexity:** the project integrates multiple live data sources (depot transaction feeds and SCADA-style sensor streams) and combines automated anonymization (k-anonymity enforcement) with a lightweight predictive-alerting model, not just a static ETL script.
- **Usability/Design:** a single executive-facing dashboard translates pipeline and compliance health into plain, operational-leadership language, so a non-technical stakeholder can understand system status at a glance.
- **Feasibility:** the architecture (ingest → clean → anonymize → ship → monitor) is built on standard, production-grade patterns — CI/CD gates, canary deploys, audit logging — that scale past the hackathon rather than a one-off notebook.
- **Real-Time Data Pipeline:** data is ingested via event-driven/streaming-style triggers from multiple simulated feeds, processed through automated quality gates, and released continuously rather than in a single static batch, with the live pipeline state shown on-screen during the demo.
- **The Actionable Insight:** the system does more than display data — it raises automated alerts when anonymization, quality, or deployment thresholds are breached, and flags at-risk records for review before they reach production.
- **A Working Demo:** a stable, interactive dashboard/app is the centerpiece of the presentation, showing live pipeline status, CI/CD health, and the executive view rather than static screenshots.
- **Business Value & Feasibility:** impact is expressed in concrete terms (analyst-hours reclaimed per week, estimated cost avoidance) alongside an honest scale-up path — real KPC feed access, stakeholder-validated numbers, and full test coverage.
- **The Pitch/Story:** the presentation is structured around Problem → Data → Solution → Actionable Outcome, so judges see the reasoning chain from KPC's pain point through to a working, adoptable tool.

3. Domain & Context

Industry: Petroleum pipeline and depot logistics operations (multi-site KPC network).

Scenario: KPC operates depot and pipeline infrastructure generating continuous transaction and sensor data across multiple sites and shifts. Several of these data streams contain fields that are sensitive under KPC's data-handling rules (staff identifiers, vehicle/truck plates, GPS/site coordinates, contractor contact details). The team is building a secure, deployable AI-tooling layer that ingests this operational data, automatically cleans and anonymizes it, and ships it through a hardened, monitored pipeline with an executive-facing view — directly addressing Problem 10 in Domain E: HSE, Compliance & Deployment Readiness.

4. Data Source

- Phase 1 (current): structurally realistic synthetic depot and SCADA-style operational logs (transaction volume, pressure/flow readings, staff and vehicle identifiers) used to build and test the ingestion, cleaning, and anonymization pipeline.
- Phase 2 (next): where available, real or real-shaped depot/maintenance data from the sponsor organization to validate the anonymization policy and quality gates against genuine operational messiness; if unavailable, a public operations or logistics dataset with comparable structure will be used.
- Fallback: continued use of the calibrated synthetic feed with injected data-quality issues (duplicates, missing fields, out-of-range sensor values) and synthetic sensitive-field patterns, if neither real nor suitably public data can be secured in time.

5. Success Metrics

- **Anonymization coverage:** 100% of identified sensitive fields masked before any record reaches analytics, enforcing $k\text{-anonymity} \geq 5$.

- **Data quality gate pass rate:** maintain at or above 95% across automated schema, range, duplicate, and completeness checks.
- **Alert / false-positive rate:** reduce noisy or low-value automated alerts to below 10% of total alerts raised.
- **Deployment reliability:** ship through a gated CI/CD pipeline with canary release and automatic rollback, targeting $\geq 99.5\%$ service uptime.
- **Business case:** produce a quantified analyst-hours-saved / cost-avoidance estimate, reviewed and sanity-checked with an Em-Tech or KPC stakeholder before Stage 3.

6. Initial Timeline — Weeks 7-12

- **Week 7 — Data sourcing & scoping:** confirm data source(s), refine the problem statement against Problem 10, and define the data-access and sensitivity-classification plan.
- **Week 8 — Real-time ingestion pipeline:** build the event-driven/streaming ingestion layer across depot and SCADA-style feeds with initial data-quality gates.
- **Week 9 — Anonymization engine:** implement automated field detection, masking, and k-anonymity enforcement, with full audit-trail logging per record.
- **Week 10 — CI/CD hardening & alerting model:** stand up the tested, gated deployment pipeline (build, scan, canary, monitor) and train the first predictive-alerting model.
- **Week 11 — Executive dashboard & validation:** finalize the executive-facing view, backtest alerting performance, and build the ROI / cost-avoidance case with stakeholder review.
- **Week 12 — Final delivery:** final hardened prototype, technical and executive documentation, and the live demo/pitch presentation.